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English



Due Jan 18, 11:59 PM GT

Instructions

Module 8 Summative Assessment

1. What is the role of z in a mixture model?
- Multiple Choice Questions**
- ☐ It represents the mean of the normal distribution.
 - ☒ It is the indicator variable that assigns data points to different components.
 - ☐ It is the parameter that defines the variance of the mixture components.
 - ☐ It is the parameter that defines the mean of the mixture components.
2. What is a limitation for the label switching problem in mixture models?
- Multiple Choice Questions**
- ☐ Reducing the number of mixture components.
 - ☐ Averaging the parameters across the components.
 - ☒ Using information prior to the model as an ordering on the labels.
 - ☐ Ignoring the issue entirely.
3. In the context of finite mixture models, what is the role of observed indicators?
- Multiple Choice Questions**
- ☐ They simplify the hierarchical model.
 - ☐ They are used to estimate the variance of the model.
 - ☒ They specify the mixture component from which each observation is drawn.
 - ☐ They represent the observed data.
4. What is a common problem when choosing the number of mixture components, K , in a finite mixture model?
- Multiple Choice Questions**
- ☒ Overfitting, where the model is too closely tailored to the training data.
 - ☐ Underfitting, where the model is too simple to capture the underlying structure.
 - ☐ It simplifies the computation.
 - ☐ It always reduces model flexibility.
5. Which of these financial applications would warrant the use of mixture models?
- Multiple Choice Questions**
- ☐ To reduce the number of financial assets in a portfolio.
 - ☐ To ensure all assets have the same return rate.
 - ☐ To identify single-risk profiles for all investors.
 - ☒ To classify financial assets into different risk categories.
6. Based on what you know about mixture models, which could be a potential application in image processing?
- Multiple Choice Questions**
- ☐ To increase the resolution of images.
 - ☐ To reduce the size of image files.
 - ☐ To convert images to grayscale.
 - ☒ To classify different regions or objects within an image.
7. What distribution is commonly assumed for the vector of mixture indicators in a finite mixture model?
- Multiple Choice Questions**
- ☐ Exponential distribution.
 - ☒ Multinomial distribution.
 - ☐ Normal distribution.
 - ☐ Poisson distribution.
8. How might mixture models be used in environmental science?
- Multiple Choice Questions**
- ☐ To combine all environmental data into one model.
 - ☒ To identify distinct pollution sources.
 - ☐ To simplify the climate data into a global model.
 - ☐ To eliminate the need for field studies and weather balloons.
9. What is the primary purpose of using finite mixture models?
- Multiple Choice Questions**
- ☒ To model data arising from multiple subpopulations.
 - ☐ To simplify the data analysis process.
 - ☐ To eliminate the need for hierarchical modeling.
 - ☐ To ensure all data follows a single normal distribution.
10. Based on what you know, how could mixture models assist in the analysis of social network data?
- Multiple Choice Questions**
- ☒ By identifying distinct communities or clusters within the network.
 - ☐ By reducing the number of users in the network.
 - ☐ By simplifying the social network structure.
 - ☐ By grouping all users into one community.
11. Which of the following is a valid method for selecting the number of mixture components in a model?
- Multiple Choice Questions**
- ☒ Using the Bayesian Information Criterion (BIC) or Bayesian Information Criterion (BIC) for model selection.
 - ☐ None of the above.
 - ☐ Arbitrarily choosing the number based on convenience.
 - ☐ Selecting the number based on the largest possible number of components that the software can handle.
12. What is the label switching problem in mixture models?
- Multiple Choice Questions**
- ☒ The invariance of the likelihood function to permutations of the component labels, leading to non-identifiable parameters.
 - ☐ The necessity of re-estimating parameters when new data is added to the model.
 - ☐ The inability to differentiate between different mixture components due to identical labels.
 - ☐ The occurrence of components being assigned the same data points repeatedly.
13. What does the term 'posterior predictive distribution' refer to in Bayesian mixture models?
- Multiple Choice Questions**
- ☐ The likelihood function.
 - ☐ The distribution of observed data.
 - ☒ The distribution of future observations based on the posterior distribution of parameters.
 - ☐ The prior distribution of parameters.
14. What is the EM algorithm?
- Multiple Choice Questions**
- ☐ An algorithm that performs maximum expectation under parameter constraints.
 - ☐ A posterior sampling algorithm designed for Gaussian mixture models.
 - ☒ A variant of the Gibbs sampler used for hierarchical models.
 - ☐ A projection algorithm used to compute coefficients in linear models.
15. Question
- Multiple Choice Questions**
- ☐ Question D
 - ☐ Question B
 - ☒ Question A
 - ☐ Question C